Part 17 — The last two gaps: model curation, and real instrument data

I said these two were better done at a terminal or a bench than on paper. That was half right: what I could not do was *fetch* real data. What I can do — and have done here — is generate instrument data with the answer built in, so you process genuine files and then check whether your method recovered the truth. Five datasets are attached as CSV. The curation problems are computed against the core model from part 10, with errors deliberately injected.

The data files

File	Size	Problem	What it is
`R1_chromatogram.csv`	15 kB	R1	A 12-minute HPLC trace at 10 ms resolution: four peaks, a drifting baseline, a solvent front and detector noise
`R2_qpcr.csv`	25 kB	R2	45-cycle amplification curves, two targets × two samples × three replicates
`R2_qpcr_standard_curve.csv`	0.5 kB	R2	A five-point dilution series for each target, in duplicate
`R3_plate_growth.csv`	139 kB	R3	96 wells × 73 time points of OD_{600}, with blanks, an edge effect and noise
`R4_ms_isotopologues.csv`	0.4 kB	R4	A pyruvate isotopologue cluster, M+0 to M+3, in triplicate
`R5_offgas.csv`	2.7 kB	R5	24 h of outlet gas composition, dissolved oxygen and feed rate at 15-minute intervals
`ANSWERS_do_not_open_first.json`	13 kB	all	The true parameters for every dataset. Open it after you have an answer, not before

The data are simulated, and the statements say so. The point is not to pretend they came off an instrument but to give you a file with noise, drift, overlap and a known answer — which is the one thing a real instrument file cannot give you.

Group A — Genome-scale model curation

MC1 · extends part 10 chapter 37 · 2 h

A reaction that balances on carbon and still breaks the model

Someone has added one reaction to the 40-reaction core model: **4 CO_2 → oxaloacetate**. It is written with a correct carbon count. Growth on 10 mmol of glucose rises from 0.754 to **10.00 h^{-1}**.

1. Check the reaction's carbon balance. Does a carbon-balance audit catch it?
2. Count the electrons on each side using degrees of reduction. What is created from nothing?
3. Design the one simulation that exposes it in a single line, and predict its result.
4. Why did the error inflate growth 13-fold rather than slightly?
5. List three further checks that would have caught this class of error before any simulation.

Worked solution. The carbon balance is **exactly zero** — four carbons in as CO_2, four out as oxaloacetate — so a carbon audit **passes it**, which is the whole lesson. The electron count does not: CO_2 is fully oxidised and contributes nothing, while oxaloacetate ($C_4H_4O_5$) carries $4\times4 + 4\times1 - 5\times2 =$ **10 electrons**, all of them created from nothing on every turn. Because the reaction also needs no ATP, it is simultaneously a free-reductant and a free-carbon generator.

The exposing simulation is one line: **set the glucose uptake bound to zero and optimise**. A correct model returns zero growth; this one still grows, because it can build biomass out of CO_2 and nothing else. That test — "can the model grow on no carbon source?" — belongs in every model's test suite, and it is the first thing to run after any edit.

The inflation is 13-fold rather than marginal because the injected reaction relieves *every* binding constraint at once: it supplies carbon, reductant and a four-carbon precursor, so the optimiser pushes it to 406 mmol·gDW^{-1}·h^{-1} and growth runs to whatever other bound it hits. An error that relaxes the binding constraint always shows up as a large number, which is why an implausibly good result is itself a diagnostic. The three further checks: a full **elemental balance** including H and O rather than carbon alone; a **charge balance**; and the **stoichiometric consistency** test of part 10 chapter 37, which asks for a positive mass vector satisfying $S^{T}m = 0$ and fails when any reaction creates mass.

What it teaches: carbon balance is necessary and nowhere near sufficient; the growth test on an empty medium is the cheapest and most powerful curation check there is.

MC2 · extends part 10 chapter 36 · 1.5 h

Separate the errors that change the answer from the errors that change the map

The same core model contains succinate dehydrogenase and fumarate reductase, which together form a closed loop. You want to know whether to care.

1. Compute the growth rate with the loop present and with fumarate reductase blocked. What changes?
2. At 99% of optimal growth, maximise the succinate dehydrogenase flux. What value do you get and why?
3. Classify: does this error affect the objective, the flux map, or both?
4. Give the order in which you would fix a model's errors, and justify it by this result.

Worked solution. Growth is **0.7541 h^{-1} either way** — identical to ten decimal places, because the loop has no net conversion and so neither consumes nor produces anything biomass needs. Maximising the flux through it at 99% of optimum returns **1000**, which is not a biological number at all: it is the arbitrary upper bound in the model file, and its appearance in an answer is the signature of a loop.

The classification matters: this error affects **the flux map and not the objective**. So every yield, every ceiling and every deletion prediction from this model is unaffected, while every statement of the form "flux goes this way through that branch" is unreliable. That gives the repair order: **first, errors that change the objective** — mass and charge imbalances, free-energy generators, wrong reversibility that opens an impossible route, because these corrupt the numbers you quote; **second, errors that change the map** — loops and alternative optima, because they corrupt the interpretation; **third, metadata** — annotations, identifiers, SBML conformance, which affect reusability and not correctness. A group that starts with annotations because they are easy has fixed the least important category.

What it teaches: not all model errors are equal, and knowing whether an error moves the objective or only the flux map tells you whether to stop the analysis or just stop interpreting the fluxes.

MC3 · extends part 10 chapter 37 · 2 h

A reversibility error, and the unbounded ATP it buys

Acetate kinase in the model is irreversible, as written. A curator "fixes" it to be reversible — the enzyme is reversible in vitro, after all — and the model is also given an acetate uptake exchange.

1. With acetate uptake allowed and the kinase irreversible, maximise ATP maintenance flux at zero growth.
2. Repeat with the kinase made reversible.
3. Now remove every carbon source. What do you get, and what does the answer tell you?
4. Explain why "the enzyme is reversible in vitro" is not sufficient grounds for the edit.

Worked solution. With acetate uptake and the irreversible kinase, maximum ATP production is **320 mmol·gDW^{-1}·h^{-1}** — a high but finite number set by the acetate uptake bound and the respiratory chain. With the kinase reversible it becomes **1000**, the bound again: the model has acquired a cycle that makes ATP without limit. Removing all carbon sources returns **zero**, so this is not a pure free-energy generator — it needs acetate to run — which is why it would survive the M1 test and has to be caught by the ATP test instead: **maximise ATP with the biomass reaction switched off and compare with the substrate's theoretical maximum**, which for acetate is about 10 ATP per molecule.

The in vitro argument fails because reversibility in a test tube is a statement about the enzyme at whatever concentrations the assay used, while reversibility in a model is a statement about the *direction the reaction can carry net flux at cellular concentrations*. Acetate kinase operates far from equilibrium in vivo — the cell holds a high ATP-to-ADP ratio and a low acetyl phosphate pool — so the reverse direction is thermodynamically shut even though the protein catalyses it. That is the part 2 chapter 14 distinction between $\Delta G^{\circ\prime}$ and $\Delta G'$, and it is why directionality should come from the thermodynamics at physiological concentrations rather than from an enzyme's catalytic competence.

What it teaches: reversibility in a model is a thermodynamic claim about cellular conditions, and the ATP-maximisation test is what catches a wrong one.

`MC4` · extends part 10 chapter 37 · 1.5 h

A biomass coefficient error, and why its direction decides whether you notice

The core biomass reaction charges 59.81 mmol of ATP per gram dry weight. Three variants are tested, with the coefficient multiplied by 0.5, 2 and 10.

1. Compute growth for each variant.
2. Which errors are visible in the growth rate and which are not? Explain.
3. What does this imply about a biomass equation copied from a related organism?
4. Which predictions are most sensitive to a biomass error, and which are robust?

ATP coefficient	Growth (h^{-1})	Fraction of correct
29.9 (half)	0.7541	**100%**
59.81 (correct)	0.7541	100%
119.6 (double)	0.6669	88%
598.1 (ten-fold)	0.3300	44%

Worked solution. Halving the coefficient changes the growth rate **not at all**, because at the correct value the model is carbon-limited rather than ATP-limited — the slack in the ATP budget absorbs the error completely. Doubling it costs 12% and a tenfold error costs 56%. So **an error that makes biomass cheaper is invisible** in the growth rate while an error that makes it dearer is visible, and only once it is large enough to make ATP the binding constraint.

For a copied biomass equation that is an uncomfortable asymmetry: the usual direction of error, borrowing coefficients from a better-studied relative, can sit undetected. The predictions sort accordingly. Robust: **carbon-limited yield ceilings**, which depend on the precursor demands rather than the ATP term. Sensitive: **anything ATP-limited** — maintenance behaviour, growth on poor substrates, the P/O comparisons of part 14 chapter B4 — and above all **gene essentiality**, since essentiality often turns on whether one specific precursor can be made at all, so a wrong coefficient for a minor component makes its pathway spuriously essential or spuriously dispensable.

What it teaches: a biomass error is only visible when it makes the constrained resource more expensive, so the direction of the error decides whether you will ever find it.

`MC5` · extends part 10 chapter 37 · 1 h

How much of this is there in a real model?

1. Your core model has 40 reactions and the injected error was found by two checks. Scale that to a 1,400-reaction automated draft with a reported 28% of reactions unbalanced, and to a 2,583-reaction curated model at 5%.
2. How many reactions need inspection in each case?
3. Which of MC1 to MC4's checks can be automated, and which cannot?
4. Write the test suite you would attach to your own model, in order.

Worked solution. At those rates a draft has **392 reactions** to inspect and a curated genome-scale model still has **129**. That is weeks of work in the first case, which is the honest reason automated drafts are used for capability questions and not for gene-level predictions (part 10 chapter 37). Automatable: elemental and charge balance, stoichiometric consistency, blocked-reaction detection, the no-carbon growth test, the ATP-maximisation test, and loop detection by flux variability — all of these are arithmetic on the stoichiometric matrix and need no judgement. **Not** automatable: deciding whether a reaction should exist in this organism, choosing a direction when the thermodynamics are marginal, and setting biomass coefficients — each needs evidence rather than computation.

The test suite, in the order to run it: (1) does the model grow on the defined medium, and *not* grow with no carbon source; (2) is maximum ATP at zero growth within the substrate's theoretical yield; (3) is every internal reaction elementally and charge balanced; (4) is the model stoichiometrically consistent, with any deliberate currency lumping documented; (5) which reactions are blocked, and does each belong to a pathway with a missing half; (6) which reactions appear at the arbitrary flux bound in variability analysis. Run as a script on every commit, this takes seconds and catches the entire class of error in MC1 to MC3.

What it teaches: curation splits cleanly into checks a script should run on every edit and judgements only evidence can settle, and conflating the two is why drafts stay uncurated.

Group B — Real instrument data, with the answer withheld

R1 · file `R1_chromatogram.csv` · 2.5 h

Integrate a chromatogram, including the peak that overlaps your product

A 12-minute trace sampled at 10 ms. Four peaks: an impurity at 2.41 min, the product at 3.86, a co-eluting compound at 4.12 and an internal standard at 6.73. The baseline drifts upward and there is a solvent front near 1.1 min. The response factor is 0.812 g·L^{-1} per unit of product-to-standard area ratio.

1. Fit and subtract a baseline using only regions you judge to be peak-free. Say how you chose them.
2. Integrate the product valley-to-valley, then over a window wide enough to include the co-eluting peak. Report both areas.
3. Compute the product-to-standard ratio and the concentration from each integration.
4. Deconvolve the 3.86/4.12 pair by fitting two exponentially modified Gaussians, and compare.
5. Open the answer file. By how much was each method wrong, and which error would you rather have?

Expected result. Integrating valley-to-valley **clips the product's tail and reads about 5% low**; a window wide enough to include the co-eluting peak reads about **23% high**. Both are wrong, and in opposite directions, from the same data. The deconvolution should land within a couple of per cent because the peak shapes are genuinely exponentially modified Gaussians — which is the case where fitting beats integration.

Two further points the data will make for you. The internal standard is recovered almost exactly by any sensible method, because it is baseline-resolved — so a **ratio is only as good as the worse of its two integrations**. And the answer you prefer depends on use: for a process decision a consistent 5% low bias is more useful than a variable high bias, because it is correctable by the calibration; for a purity claim neither is acceptable and the method must be changed to resolve the pair.

What it teaches: integration limits are a judgement that changes the answer by tens of per cent, and a co-eluting peak is a chromatography problem that no amount of arithmetic repairs.

R2 · files `R2_qpcr.csv`, `R2_qpcr_standard_curve.csv` · 2 h

Take a qPCR run from raw fluorescence to a fold change you can defend

Two targets, a pathway gene and a reference gene, in a control and an engineered strain, three replicates each over 45 cycles. A separate five-point dilution series in duplicate gives each target's efficiency.

1. Subtract the baseline and determine C_q for all twelve curves by a fixed threshold. Report means and spreads.
2. Fit each standard curve and convert the slope to an amplification efficiency.
3. Compute the fold change by the $2^{-\Delta\Delta Cq}$ method and by the Pfaffl method using the measured efficiencies.
4. Which is larger, by how much, and which would you report?
5. Open the answer file and check your C_q values and efficiencies.

Expected result. The threshold method recovers all four C_q values to within about 0.1 cycle, giving a $\Delta\Delta C_q$ near **3.07** against a true 3.00. The standard curves give slopes near **−3.44 and −3.34**, which convert to efficiencies of **1.95 and 1.99** — that is, 95% and 99%, not the 100% that $2^{-\Delta\Delta Cq}$ assumes for both.

The two methods then disagree: **$2^{-\Delta\Delta Cq}$ gives about 8.4-fold and Pfaffl about 7.8-fold**, so the simple method **overstates by roughly 8%** here. Report the Pfaffl figure with both efficiencies quoted, because the overstatement grows with ΔC_q and with the efficiency gap — at a ten-cycle difference and the same efficiencies it would be more than 20%. The deeper point is that the efficiencies must be measured on the same assay and matrix, not taken from the primer design software, and a target below about 90% efficiency should be redesigned rather than corrected for.

What it teaches: $2^{-\Delta\Delta Cq}$ assumes perfect, equal efficiency for both targets, and the standard curve is what tells you how much that assumption costs.

R3 · file `R3_plate_growth.csv` · 2.5 h

96 wells, a blank column, an edge effect and a growth rate

Seventy-three time points over 18 hours for every well of a 96-well plate. Column 1 contains medium blanks. There is an

edge effect, and the plate has not been randomised.

1. Build the blank correction from column 1, including its drift, and apply it.
2. Fit a Gompertz model to every well and extract μ_{max}, lag and plateau.
3. Compare the distribution of μ_{max} and of the plateau between edge and inner wells. Which parameter carries the edge effect?
4. Rank the wells by plateau, then by μ_{max}. How much does the ranking change, and which ranking would you trust?
5. Open the answer file. How close were your fitted rates, and did you correctly identify what the edge did?

Expected result. The blanks drift upward by about 0.0008 OD per hour — roughly 0.014 over the run, which is a third of the blank value itself — so a single initial blank subtraction leaves a systematic positive drift in every well. Gompertz fitting on properly corrected data should recover μ_{max} to **within about 1%** of truth, which you can verify well by well.

The diagnostic question is the third. The edge effect in this dataset is applied to the **plateau** (an 18% amplitude increase from evaporation concentrating the culture) and **not to the growth rate**. So ranking wells by final OD puts edge wells at the top and is confounded; ranking by μ_{max} is nearly unaffected. That is the generalisable lesson — **choose the parameter that the artefact does not touch** — and it is more powerful than correcting for the artefact, because it needs no model of the artefact at all. Where you must use the plateau, estimate the edge factor from the data as in part 9 chapter 34 and note that this plate was not randomised, so the correction rests on the assumption that edge and inner wells hold comparable strains.

What it teaches: an artefact usually affects one fitted parameter and not another, so the robust response is to choose your readout rather than to correct your data.

R4 · file `R4_ms_isotopologues.csv` · 1.5 h

From a mass spectrum to a flux split

The M+0 to M+3 intensities of a three-carbon pyruvate fragment, in triplicate, from a culture fed [1-^{13}C] glucose. Natural ^{13}C abundance is 1.07%.

1. Normalise the cluster and report the raw M+1 fraction.
2. Build the natural-abundance correction matrix for three carbons and invert it.
3. Convert the corrected M+1 to the oxidative pentose phosphate fraction using the relation from problem C1.
4. Compute how wrong you would have been without the correction.
5. Open the answer file and compare.

Expected result. The raw M+1 is about **0.351**; correction brings it to **0.337**, which converts to an oxidative fraction of **0.326** against a true **0.32** — recovered to within the 2% measurement noise. Skipping the correction would have given an oxidative fraction about **0.028 lower**, a relative error near 9% on the flux you are trying to measure, and the error is systematic rather than random, so replicates will not reveal it.

Two points worth drawing out. The correction's size scales with the number of carbons in the fragment, so a larger fragment needs a larger correction and the matrix is no longer nearly diagonal — a six-carbon fragment has a natural M+1 near 6.4%. And the correction must also account for any heavy isotopes in a derivatisation group, which is why silylated samples need a correction matrix built from the full elemental formula of the fragment rather than from its carbons alone.

What it teaches: natural-abundance correction is a small, systematic, mandatory step whose omission biases a flux in a direction replicates cannot expose.

R5 · file `R5_offgas.csv` · 2 h

Read a fermentation from its off-gas alone

Twenty-four hours of outlet oxygen and carbon dioxide, dissolved oxygen and feed rate, every 15 minutes, from a 100 m³ vessel at 0.5 vvm. Inlet air is 20.95% O_2 and 0.04% CO_2.

1. Compute OUR, CER and RQ at every time point using an inert balance for the outlet flow.
2. Identify the feed step from the data alone, and then the time at which the physiology changes.
3. Quantify the RQ before and after, and say what the change means.
4. Reconstruct the specific oxygen uptake rate before and after, given that biomass plateaus near 45 gDW/L.
5. What would you have done, and when, if you had been watching this run live? Open the answer file.

Expected result. OUR rises exponentially and plateaus near **180 mmol·L⁻¹·h⁻¹**, which is the vessel's transfer ceiling rather than the culture's demand. RQ sits at **1.00 before hour 14 and 1.18 after**, and the feed step is at hour 12 — so the physiological change **lags the feed step by about two hours**, which is the interval in which an operator can still act. The specific uptake rate rises from **4.0 to 5.6 mmol·gDW⁻¹·h⁻¹**, meaning the cells were asked for 40% more oxygen than before.

The live reading is the point of the problem. An RQ above 1 with a feed increase and a dissolved oxygen trace that has stopped falling means the culture is fermenting part of its carbon — overflow, as in part 1 chapter 6 — because the oxygen it would need is no longer available. The action is to **reverse the feed step**, not to raise the stirrer, since OUR has already plateaued at the transfer limit and more agitation buys little. Waiting for the acetate assay to come back from the laboratory costs hours during which carbon is being converted to by-product, which is why the off-gas analyser earns its cost.

What it teaches: off-gas gives you metabolism in real time, and the lag between an operating change and its physiological consequence is the window in which the run can still be saved.

Closing notes to part 17

How to work these

Do the R problems with the answer file closed. Write your analysis, record your number, *then* open `ANSWERS_do_not_open_first.json` — it holds the true growth rates, C_q values, efficiencies, per-well parameters, labelling fraction and RQ shift. Checking afterwards is how you learn what your method costs; checking first turns the exercise into a reading comprehension test.

The MC problems need COBRApy or any LP solver and the core model, either from BiGG or rebuilt from part 10's listing. They are stated so that the *diagnosis* is the work: each injected error has a signature, and the skill is knowing which test exposes which signature.

What the ten establish

- **MC1** — a carbon-balanced reaction inflated growth 13-fold, and the one-line test that exposes it is growth with no carbon source.
- **MC2** — a loop changes the flux map and leaves the objective identical to ten decimal places, which sets the repair order.
- **MC3** — one reversibility edit took maximum ATP from 320 to the arbitrary bound.
- **MC4** — halving the biomass ATP cost is invisible, doubling it costs 12%; the direction of the error decides whether you find it.
- **MC5** — 392 reactions to inspect in a draft, 129 in a curated model; which checks automate and which cannot.
- **R1** — the same chromatogram reads 5% low or 23% high depending on where you put the integration limits.
- **R2** — $2^{-\Delta\Delta Cq}$ overstates by 8% at measured efficiencies of 95% and 99%.
- **R3** — the edge effect lands on the plateau and not on μ_{max}, so the robust answer is to change readout, not to correct.
- **R4** — omitting natural-abundance correction biases the flux split by 9%, systematically.
- **R5** — physiology changes two hours after the feed step, and that lag is the operator's window.

Where the whole problem set now stands

Source	Problems	Data
Your curriculum appendix, A1–W3	54	Stated inline
Part 15, X1–X24	24	Computed here
Part 15, Y1–Y6	6	Real public datasets, not run
Part 16, Z1–Z9	9	Computed here
Part 17, MC1–MC5	5	Computed here against the part 10 model
Part 17, R1–R5	5	**Data files supplied**, truth withheld in a separate file
Total	**103**	

I verified that each of the five datasets returns its own truth when processed correctly, which is how I know the problems are solvable: the chromatogram's internal standard recovers to 0.1%, the qPCR C_q values to 0.1 cycle, the Gompertz fits to about 1%, the labelling fraction to 0.006, and the RQ shift exactly. Where your answer differs from the file by more than that, the difference is in your method, which is the whole point.

What genuinely remains out of reach on paper: operating an instrument, judging a chromatogram's suitability by eye, and anything needing a strain in a flask. Those are the bench, and no document substitutes.